

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250276252

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Goldmeier; Steven Neil

SELF-PROPELLED NOVELTY TOY

Abstract

A self-propelled toy includes a toy body, a drive and control module, including a housing, an axle with wheels arranged on opposing ends and an axel drive/driven gear attached to the drive and control module within the housing. A pull-back mechanism arranged in, on or otherwise attached and mechanically connected to the drive and control module Upon pulling back the toy body while maintaining the wheels touching a ground like surface, the pull-back mechanism generates and stores mechanical potential energy in a spring/tensioning motor mechanism, which upon release of the toy body, translates to kinetic energy and with the wheels touching the ground like surface drives the toy body in a direction opposite that of the pull-back direction.

Inventors: Goldmeier; Steven Neil (Plainview, NY)

Applicant: Goldmeier; Steven Neil (Plainview, NY)

Family ID: 96881728

Appl. No.: 19/192252

Filed: April 28, 2025

Related U.S. Application Data

parent US continuation-in-part 18124322 20230321 PENDING child US 19192252

Publication Classification

Int. Cl.: A63H11/10 (20060101); A63H3/38 (20060101); A63H5/00 (20060101); A63H29/02 (20060101); A63H31/08 (20060101)

U.S. Cl.:

CPC A63H11/10 (20130101); A63H3/38 (20130101); A63H5/00 (20130101); A63H29/02

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation in part (CIP) application of application Ser. No. 18/124,322, filed Mar. 21, 2023 (“the parent application”), which parent application claims priority under 35 USC § 119(e) from U.S. Provisional Patent Application No. 63/223,892 filed Mar. 25, 2022 (“the provisional patent application”); the contents of each of the parent application and the provisional patent application are incorporated herein by reference.

BACKGROUND OF THE INVENTION

[0002] The present invention relates to wheeled, self-propelled toys, and more particularly related to a self-propelled novelty toy configured with a pull-back mechanism and inter alia elements that compel the toy either to “scream” or “flash eyes” or both when the self-propelled toy is in a primed (wound), energized with stored potential energy state and in a “go” (moving), energy releasing (kinetic) state.

SUMMARY OF THE INVENTION

[0003] The invention overcomes the shortcomings of the known self-propelled toys with pull-back mechanisms.

[0004] In an embodiment, the invention provides a self-propelled toy. The toy includes a toy body, a “drive and control” module, including a housing, a connector for connecting the drive and control module to, and disconnecting the drive and control module from, the toy body, an axle with wheels on opposing ends and an axel drive/driven gear, attached to the drive and control module housing, a pull-back mechanism arranged in or on the drive and control module housing that includes a motor drive/driven gear that drivingly connects to the axel drive/driven gear, wherein upon pulling back the toy body with the wheels touching the ground, the pull-back mechanism generates and stores mechanical potential energy, which upon release of the toy body, translates to kinetic energy to drive the toy in a direction opposite that of the pull-back direction and a controller arranged on or in the drive and control module housing, connected to the pull-back mechanism and electrically connected to at least one light emitting diode (LED) in the toy body, the controller including a microcontroller, or other logical elements and an rechargeable energy source for powering the microcontroller or other logical elements, and the at least one LED.

[0005] The controller includes at least one sensing element for sensing when the pull-back mechanism has generated and stored mechanical energy, defining a “primed” logical state, and when toy is released to allow the toy to move forward, defining a “go” state. If in the primed state and the go state, the at least one LED is powered to generate a light signal. Preferably, the toy body is any of a mammal, a reptile, a fish, a dinosaur, other animals/insects, or a human sports player, all with an eye, wherein the at least one LEDs are in the eyes. The toy may also include a sound transducer electrically connected to the controller that emits a sound signal when in the primed state and the go state. The toy body may embody a mammal, a reptile, a fish, a dinosaur, other animals/insects, or a human sports player, all with mouths, where a sound transducer is arranged in the mouths. The sound transducers emit a sound signal when in the primed state and the go state.

[0006] The mammals, reptiles, fish, dinosaurs, other animals/insects, and sports players, move with the pull-back mechanism. While the mammals, reptiles, fish, dinosaurs, and other animals/insects, generally move in a forward direction, but the sports players move in a forward direction, but preferably in a sliding forward direction. For example, when the sports player is a soccer player, the player slides with his/her knees bent in a forward direction, with the arms out stretched in a celebratory pose. Other sports figures such as baseball/softball players can slide feet first towards a

base or Homeplate.

[0007] Additionally, track and field runners can be depicted as upright and running forwards. Furthermore, skiers can be depicted as sliding on skis down a ski slope. Other moving sports figures may include basketball players, football players, hockey players, tennis players, or pickleball players. Optionally, the sliding or moving character can be depicted upon a small portion of a playing field, so that the pull-back mechanisms are beneath the small portion of the playing field, which will advance forward with the sports player positioned above the playing field.

[0008] In an embodiment, the self-propelled toy includes a toy body, a drive and control module, including a housing; an axle with wheels arranged on opposing ends and an axle drive/driven gear attached to the drive and control module within the housing, a pull-back mechanism arranged in, on or otherwise attached and mechanically connected to the drive and control module, which pull-back mechanism includes a motor drive/driven gear, wherein upon pulling back the toy body while maintaining the wheels touching a ground like surface, the pull-back mechanism generates and stores mechanical potential energy in a spring/tensioning motor mechanism, which upon release of the toy body, translates to kinetic energy and with the wheels touching the ground like surface drives the toy body in a direction opposite that of the pull-back direction; and a controller arranged on or in the drive and control module housing, connected to the pull-back mechanism and electrically connected to at least one light emitting diode (LED) and at least one sound transducer arranged in the toy body, the controller including an energy source for powering the controller, the at least one LED and the at least one sound transducer.

[0009] The toy body may be any of a mammal, a reptile, a fish, a dinosaur and a sliding or running human sports player, all with a plurality of eyes, and wherein said at least one LED is arranged in each of said plurality of eyes. The sound transducer is electrically connected to the controller and is controlled by the controller to emit a sound signal when the pull-back mechanism is in a primed state, in a go state or in both states. The sound transducer emits a sound signal when the pull-back mechanism is in the primed state and the go state. The toy body may comprise a sliding or running human sports player, which is selected from the group of consisting of a sliding soccer/baseball/softball player, a runner, a skier, a basketball player, a football player, a hockey player, a tennis player and a pickleball player.

[0010] The toy may further comprise at least one sensing element for sensing when the pull-back mechanism has generated and stored mechanical energy and/or is releasing stored energy to drive the toy. Detected generated stored mechanical energy defines a “primed” logical state, and when toy is released to allow the toy to move forward through use of the stored mechanical energy defines a go state. Preferably, the LED is activated by a user touch, or when the toy is in the primed state, or when the toy is in the go state or any combination thereof. The sound generating device is included and is activated by a user touch, The sound generating device is included and is activated when the toy is in the primed state. or when the toy is in the primed state, or when the toy is in the go state or any combination thereof. The sound generating device is included and is activated by a user touch.

Description

BRIEF DESCRIPTION OF THE DRAWING FIGURES

[0011] Further features and advantages of the invention will become apparent from the description of embodiments that follows, with reference to the attached figures, wherein:

[0012] FIG. 1 depicts a child holding a phantom 2-wheel triceratops, with a full line triceratops in full line adjacent and rolling away from the child;

[0013] FIG. 2 is an enlarged view off the Triceratops of FIG. 1, inside elevation;

[0014] FIG. 3 is an underside perspective view of the triceratops with the mechanism exploded

down from the mount;

[0015] FIG. 4 shows the wheel mechanism removed from the housing:

[0016] FIG. 5 is a side elevation of the wheel mechanism, attempting to show the gear setup;

[0017] FIG. 6A is a side view of the toy highlighting the connection from a controller to drive LED eyes;

[0018] FIG. 6B is a side view of the toy highlighting the connection from a controller to drive the sound transducer in the mouth;

[0019] FIG. 7 is a perspective view of a celebratory soccer sports player sliding with the knees forward and the arms outstretched; and,

[0020] FIG. 8 is a perspective view of the celebratory soccer sports player shown in partial cutaway view, displaying interior mechanical and electrical components for the pull-back mechanism and sound and light producing components.

DETAILED DESCRIPTION OF THE INVENTION

[0021] The following is a detailed description of exemplary embodiments of the invention, which are depicted in the accompanying drawings. The exemplary embodiments are presented in such detail as to clearly communicate the invention and are designed to teach how to make and use these exemplary embodiments to a person of ordinary skill in the art. However, the amount of detail offered is not intended to limit the contemplated variations of the embodiments of the disclosed invention. On the contrary, the inventor intends to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the disclosed invention, as defined by the appended claims.

[0022] In an embodiment, the invention provides a self-propelled toy **10** (FIG. 1). As shown in FIGS. 2-5, the self-propelled toy comprises a toy body **20** attached to a drive and control module **42**. The drive and control module **42** includes a housing **42**, a dricon-connect portion **34** for connecting the drive and control module to, and disconnecting the drive and control module from, the toy body (See body connect portion **32**), includes an axle **46**, wheels **48** on opposing ends and an axel drive/driven gear **50**, a pull-back mechanism **60** arranged in or on a pull-back mechanism housing **61** that includes a motor drive/driven gear **62** that drivingly connects to the axel drive/driven gear, wherein upon pulling back the toy body with the wheels touching the ground, the pull-back mechanism **50** generates and stores mechanical potential energy, which upon release of the toy body, translates to kinetic energy to drive the toy in a direction opposite that of the pull-back direction.

[0023] The drive and control module **40** includes a spring/tensioning motor mechanism **64**, a motor drive/driven gear **62** and a controller **70**, all arranged on or in the drive and control module housing **61**. The controller **70** includes a controller sensor electrically connected to the pull-back mechanism for sensing a primed state and a go state. The controller **70** is electrically connected to at least one sound transducer **74** and at least one light emitting diode (LED) **75** that are arranged in the toy body **20** and electrically connected thereto via wires **72**. Preferably, the controller include a microcontroller **73**, or other logical elements and an energy source such as a battery (not shown) for powering the controller/microcontroller **70/73** or other logical elements, the at least one sound transducer **70** and the at least one LED **75**.

[0024] The controller sensor **71** senses when the spring/tensioning motor mechanism **64** of the pull-back mechanism **60** has generated and stored mechanical energy, defining a “primed” logical state, and when toy is released, utilizing the stored mechanical energy as kinetic energy to allow the toy to move forward, defining a “go” state. If in the primed state and the go state, the sound transducer **74** and the LED **75** are controlled to emit sound and light, respectively by the controller. Preferably, the toy body is any of a mammal, a reptile, a fish, a dinosaur, other animals/insects, or a human sports player, all with an eye, wherein the at least one LEDs are in the eye. Preferably, the sound transducer is controlled by the controller to emit a sound signal when transitioning from the primed state to the go state. The sound transducer is preferably arranged in or proximate a mouth of the

mammal, a reptile, a fish, a dinosaur, other animals/insects, or a human sports player, comprising the toy body.

[0025] A mechanical connector **30** is arranged in the toy body with a body-connect portion **32** (e.g., preferably underneath the toy body to minimize its visibility). The body-connect portion **32** of the mechanical connector **30** includes body-connect pins **31** for connecting the connector **30** to the drive and control module housing **42**, i.e., to the drive and control module **40**.

[0026] The dricon-connect portion **34**, which is part of the housing **42** connects to the toy body **20**, i.e., the body-connect portion **32** of the mechanical connector **30**. Connect pins **31** on the body connect portion **32** of the mechanical connector **30** are received in openings for connect pins **35** arranged in an upper side of the dricon-connect portion to connect the body **20** to the housing **42**. The housing **42** includes a pull-back mechanism **60** and a controller **70** (FIG. 4), as explained above.

[0027] The pull-back mechanism **60** is seated in a cutout of outer side walls of the dricon-connect portion **34**. The pull-back mechanism includes a pull-back mechanism housing **61**. Extending out of the sides of the pull-back mechanism housing **61** is a single axle **46** with wheels **48** attached to the axle on opposing ends. An axel drive/driven gear **50** is arranged to drive the wheels when the toy **10** is released in its go state after the toy has been would in its primed state. The wheels contact the ground during intended use, where the housing **42** also includes a housing ground contact point **44**. The housing ground contact point **44** is arranged relatively towards the front of the housing, and the lower side of the housing. The wheels **48** and the ground contact point **44**, arranged on an underside of the housing **42** between the wheels **48** and a front end of the housing **42** therefore provide for 3 point balanced ground contact. Such arrangement allows for the toy **10** to sit on a ground surface it is used on and/or move stably with minimal friction between the ground contact point **44** and the ground surface.

[0028] The pull-back mechanism **60** preferably comprises a spring/tensioning motor mechanism **64** and is arranged in or on the pull-back mechanism housing **42**. spring/tensioning motor mechanism **64** drives the motor drive/driven gear **62** and thereby the axel drive/driven gear **50**. When the toy body **20** is connected to the housing's dricon-connect portion **34**, the wheels are contacted to the ground surface and drawn back (against an internal spring/tensioning motor mechanism **64**) to generate and store mechanical potential energy, which is then released as a grip on the toy **10** is released and converted to kinetic energy (go state). The releasing energy drives the gears (e.g., the motor drive/driven gear **60**, the axle driven/drive gear **50**) to drive the wheels **40** and move the toy forward using the stored energy.

[0029] In more detail, when the potential energy is released (go state), the stored energy in the spring/tensioning motor mechanism **64** drives the motor drive/driven gear **62** which in turn drives the axel drive/driven gear **50**, and, therefore, the axel **46** and wheels **48**. The pull-back mechanism **60** and spring/tensioning motor mechanism **64** therein may be implemented by any means known to the skilled person, as shown in the figures, for example, in reliance upon a spring loaded tensioning mechanism **64**.

[0030] FIG. 6A is presented to highlights controller **70** maintained within the housing **42** of the drive and control module **40**. As earlier mentioned, the controller, which is connected to an energy storage dives such as a battery and also is electrically connected to the eyes **22** via wire(s) **72**, provides an electrical potential sufficient to light up the LEDS **75** and to cause sound to be emitted from sound transducer **74**. And as mentioned the controller **70** may include a microcontroller, or other logical elements (not shown) and the energy source (rechargeable in an embodiment, where a connector connected to the controller or the energy source directly for recharging is included at an unobtrusive location on an outer surface of the toy body **20**). Alternatively, the energy source may be an electromechanical cell or a microgenerator, which includes a micromagnet and rotor that turns with the axle to generate a current accumulated and stored as an electrical potential in an energy storage device (also not shown), as known to the skilled person.

[0031] FIG. 6B highlights a controller **70** maintained within the housing **42** that is electrically connected to the eyes **22** via wire(s) **72** to provide an electrical potential sufficient to light up the LEDS embodying same and connected via wires **72** to sound transducer **74** in (or proximate the open mouth **24** to generate sounds according to the logic implemented by the controller **70**.

[0032] The controller **70** includes is in contact with at least one sensing element (controller sensor) for sensing that the pull-back mechanism **60** has generated and stored mechanical energy, defining a “primed” logical state and that the pull-back mechanism is releasing energy in a “go” state. The microcontroller or logical elements might operate in reliance, for example, on a variable energy_store=1 that defines the primed state logically (e.g., goes to logical high state as soon as energy starts to be stored), and the release state where the wheels are driven to drive the toy forward (e.g., release_to_go=1). The sensor, or an additional sensor, senses when the energy is accumulating and being dispensed to drive the toy forward (when released by a use) with stored mechanical energy used as driving (kinetic) energy. When energy_store=1 and release_to_go=1, AND logic (energy_store AND release_to_go=1) enables electrical connection between the microcontroller and logical elements to drive the LEDs (eyes **22**) or the sound transducer **74** or both, for a limited time, intermittently, or until all the potential energy is detected to be used.

[0033] In the alternate embodiment of FIGS. 7 and 8, the sliding sports player **110** has a body **120**, which may comprise any toy body, but preferably a 3-dimensional (3D) depicted sports player in a sliding or running position, and most preferably, the toy body **120** includes eyes **122** in the form of light emitting diodes and/or a mouth **124** that is at least partially open and in which is positioned a sound-emitting element **126**, such as a transducer, where the eyes and sound-emitting element are activated to emit sound and light (in a pattern), respectively, when the self-propelled toy is pulled back or wound and released.

[0034] The drive and control module **140** embodies a housing **142**. On a bottom side of the housing (opposite the top side with the dricon-connect portion, similar to the dricon-connect portion **34** shown in FIGS. 1-6B), a single axle **146** with wheels **148** on opposing ends and an axel drive/driven gear **150** driven by axle driven/drive gear **150** are arranged. The wheels **148** contact the ground during intended use, where the housing also includes a ground contact point Not shown in FIGS. 7 and 8), which is arranged relatively towards the front, when considering that the axle **146** and wheels **148** are arranged relatively towards the back of the housing **142** of the drive and control module **140**. The wheels **148** and the ground contact point therefore provide for 3 point balance ground contact, so the toy may sit or move stably with minimal friction between the ground contact point and the ground.

[0035] A pull-back mechanism **160** (shown in brackets in FIGS. 7 and 8), which preferably comprises a spring-loaded motor mechanism, also is arranged in or on the housing of the drive and control module **140**, and includes a motor drive/driven gear **162**, which drivingly connects to the axel drive/driven gear **150**. When the toy body is connected to the housing or the pull back mechanism **160**, with pull-back motor **162**, of drive and control module **140**, is wound by drawing the toy backwards with the wheels contacting the ground or another surface, mechanical potential energy is generated (receiving mechanical energy against the spring tension) to generate and store mechanical potential energy.

[0036] When the toy is released from the user's hand, the stored energy from the pull-back motor **162** drives the axel drive/driven gear **150** to drive the motor drive/driven gear **162** (operating as a driven gear). When the toy body has potential energy, and is released, the motor drive/driven gear **162** drives the axel drive/driven gear **150**, and, therefore, the axel and wheels. The pull-back mechanism **160** may be implemented by any means known to the skilled person, for example, in reliance upon a spring or other tensioning mechanism.

[0037] FIG. 8 highlights that the controller **170** is maintained within the housing **140** and is electrically connected to the eyes **122** of the sports player **110** via wire(s) **172** to provide an electrical potential sufficient to light up the LEDS embodying same. The controller **170** preferably

includes a microcontroller, or other logical elements (not shown) and an energy source such as a battery for powering the microcontroller or other logical elements, and the eyes. The energy source may be an electromechanical cell or a microgenerator, which includes a micromagnet and rotor that turns with the axle to generate a current, which is accumulated and stored as an electrical potential in an energy storage device (also not shown), as known to the skilled person.

[0038] FIG. 8 also highlights that the controller 170 is electrically connected to a sound transducer 174 in the mouth 124 via a wire 173 to generate sounds (in addition to lighting the eyes) according to the logic of controller 170. As explained, the controller preferably includes a microcontroller, or other logical elements (not shown) and an energy source for powering the microcontroller or other logical elements, the eyes 122 and sound transducer 174. The energy source may be an electro-mechanical cell or a microgenerator, which includes a micromagnet and rotor that turns with the axle to generate a current, which is accumulated and stored as an electrical potential in an energy storage device (also not shown), as known to the skilled person.

[0039] The controller 170 includes at least at least one sensing element for sensing that the pull-back mechanism 160 has generated and stored mechanical energy, defining a “primed” logical state. The microcontroller or logical elements might implement in reliance, for example, on a variable energy_store=1 to define this primed state logically (e.g., goes to logical high state as soon as energy starts to be stored). The sensor, or an additional sensor senses when the toy is released to move forward, being driven by the stored mechanical energy used as kinetic energy. The microcontroller or logical elements might implement this “go” state in reliance, for example, on a variable release_to_go=1 to defined this release to go state logically (e.g., goes to logical high state as soon as energy is being used to drive axle and move toy forward).

[0040] When energy_store=1 and release_to_go=1, AND logic (energy_store AND release_to_go=1) enables electrical connection between the microcontroller and logical elements to drive the LEDs (eyes 122) or the sound transducer 174 or both, for a limited time, intermittently, or until all the potential energy is detected to be used.

[0041] Although the foregoing invention has been described in terms of certain specific embodiments, other embodiments of the invention will be apparent to those of ordinary skill in the art from the disclosure herein. Moreover, the described embodiments have been presented by way of example only and are not intended to limit the scope of the disclosure. Indeed, the novel processes and systems described herein may be embodied in a variety of other forms without departing from the spirit thereof.

TABLE-US-00001 REFERENCE NUMERAL ELEMENT IDENTIFIERS Self-propelled toy 10
Toy body 20 Eyes 22 Mouth 24 Sound emitting element 26 Mechanical connector 30 Body-
connect portion 32 Body side connect pins 31 Dricon connect portion 34 Openings for connect
pins 35 drive and control module 40 housing 42 housing ground contact point 44 axle 46
wheels 48 axle driven/drive gear 50 pull-back mechanism 60 pull-back mechanism housing 61
motor drive/driven gear 62 spring/tensioning mechanism 64 controller 70 controller sensor 71
wire(s) 72 microcontroller 73 Sound transducer 74 Light-emitting diode (LED) 75 Self-
Propelled Toy Sliding 110 Sports Player Toy Body 120 Sports Player's Eyes 122 Sports Player's
Mouth 124 Sound Emitting Element in Mouth 126 Drive and Control Module 140 Housing 142
Axle 146 Drive Wheels 148 Axle driven/drive gear 150 Pull-back Mechanism 160 Pull-back Motor
162 Stabilizer Wings 163, 163a Pull-back Mechanism Controller 170 Wire from Light Emitting
172 Element in Eyes Wire from Sound Emitting 173 Transducer in Mouth Sound Emitting
Transducer in Mouth 174 Knees of Sliding Player 180, 180a Extended Arms of Sliding Player 182,
182a

Claims

- 1.** A self-propelled toy, comprising: a toy body; a drive and control module, including a housing; an axle with wheels arranged on opposing ends and an axle drive/driven gear attached to the drive and control module within the housing; a pull-back mechanism arranged in, on or otherwise attached and mechanically connected to the drive and control module, which pull-back mechanism includes a motor drive/driven gear, wherein upon pulling back the toy body while maintaining the wheels touching a ground like surface, the pull-back mechanism generates and stores mechanical potential energy in a spring/tensioning motor mechanism, which upon release of the toy body, translates to kinetic energy and with the wheels touching the ground like surface drives the toy body in a direction opposite that of the pull-back direction; and a controller arranged on or in the drive and control module housing, connected to the pull-back mechanism and electrically connected to at least one light emitting diode (LED) and at least one sound transducer arranged in the toy body, the controller including an energy source for powering the controller, the at least one LED and the at least one sound transducer.
 - 2.** The toy of claim 1, wherein said toy body is any of a mammal, a reptile, a fish, a dinosaur and a sliding or running human sports player, all with a plurality of eyes, and wherein said at least one LED is arranged in each of said plurality of eyes.
 - 3.** The toy of claim 2, wherein the sound transducer is electrically connected to the controller and is controlled by the controller to emit a sound signal when the pull-back mechanism is in a primed state, in a go state or in both states.
 - 4.** The toy of claim 3, wherein the sound transducer emits a sound signal when the pull-back mechanism is in the primed state and the go state.
 - 5.** The toy of claim 4, wherein the said toy body comprises said sliding or running human sports player, which is selected from the group of consisting of a sliding soccer/baseball/softball player, a runner, a skier, a basketball player, a football player, a hockey player, a tennis player and a pickleball player.
 - 6.** The toy of claim 1, further comprising at least one sensing element for sensing when the pull-back mechanism has generated and stored mechanical energy and/or is releasing stored energy to drive the toy.
 - 7.** The toy of claim 6, wherein detected generated stored mechanical energy defines a “primed” logical state, and when toy is released to allow the toy to move forward through use of the stored mechanical energy defines a go state.
 - 8.** The toy of claim 7, wherein the LED is activated by the user.
 - 9.** The toy of claim 7, wherein the LED is activated when the toy is in the primed state.
 - 10.** The toy of claim 7, wherein the LED is activated when the toy is in the go state.
 - 11.** The toy of claim 7, wherein a sound generating device is included and is activated by a-user touch.
 - 12.** The toy of claim 7, wherein a sound generating device is included and is activated when the toy is in the primed state.
 - 13.** The toy of claim 7, wherein a sound generating device is included and is activated when the toy is in the go state.
 - 14.** The toy of claim 7, wherein a sound generating device is included and is activated when the toy is in the go state and in the primed state.
-